Composition Is Invisible and Irrelevant: Forecast Irrelevance in Feedback Markets

Draft for the authors' proof check, 14 November 2026

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Status. Not yet in the paper. Part (i) is proved in full below. Part (ii) rests on a cited theorem on autoregressive prediction. Part (iii) has a proof sketch that needs to be written out. The numerical checks are in `research_extension/irrelevance/`. Nothing here has been reviewed by anyone but the system that wrote it.

1 Setting

Use the paper's model with time-varying weights. Group $i = 1, \dots, K$ has memory rate $r_i = 1 - \eta_i \in [0, \bar{r}]$, $\bar{r} < 1$, and weight $q_{i,t} \geq 0$:

$$x_t = a\, x_{t-1} + \kappa \sum_i q_{i,t}\, y_{i,t-1} + \varepsilon_t, \qquad y_{i,t} = r_i\, y_{i,t-1} + \eta_i x_t, \qquad y_{i,-1} = 0,$$

with ε_t i.i.d., mean zero, variance σ^2, finite fourth moment, and uniform stability $a + \kappa \sum_i q_{i,t} \leq s < 1$ for all t. Two observers forecast x_{t+1}:

- the *price observer* $\mathcal{P}_t = \sigma(x_0, \dots, x_t)$;
- the *composition observer* $\mathcal{C}_t = \sigma(x_s,\, q_{i,s},\, q_{i,s} y_{i,s} :\ s \leq t,\ i \leq K)$, who sees every group's weight and position. Any class-level data (sums of positions over groups) give less than $\mathcal{C}_t$.

The *forecast value of composition* is the reduction in mean squared forecast error from $\mathcal{P}$ to $\mathcal{C}$. Write $w_{i,t} = q_{i,t}\eta_i$ and $k_{m,t} = \sum_i w_{i,t} r_i^m$ (the kernel).

2 The theorem

Theorem 1 (Composition is forecast-irrelevant)**.** *(i)* Known, constant population. *If the weights are constant and known, then*

$$\mathbb{E}[x_{t+1} \mid \mathcal{C}_t] = \mathbb{E}[x_{t+1} \mid \mathcal{P}_t] = a\, x_t + \kappa \sum_{m=0}^{t} k_m\, x_{t-m}.$$

The forecast value of composition is exactly zero.

(ii) Unknown, constant population. *If the weights are constant but unknown, the least-squares autoregression of order $L_T \asymp \log T$ fitted to $x_0, \dots, x_T$ forecasts x_{T+1} with mean squared error $\sigma^2 + O(\sigma^2 \log T/T)$. The forecast value of any composition data is therefore $O(\sigma^2 \log T/T)$. The constant depends on the population only through $\bar{r}$ and s, not on how close the memories are.*

(iii) Drifting population. *If $|q_{i,t+1} - q_{i,t}| \le v$ for all i, t, a least-squares autoregression of order $L \asymp \log(1/v)$ refitted on a trailing window of $m \asymp v^{-2/3}$ periods has excess mean squared error at most $C\sigma^2 v^{2/3} \log(1/v)$ over $\mathcal{C}$, with C depending only on $(a, \kappa, \bar{r}, s, K)$. So composition is worth at most that much for forecasting, again whatever the spacing of the memories.*

Proof of (i). From $y_{i,-1} = 0$, $y_{i,t} = \eta_i \sum_{j=0}^{t} r_i^j x_{t-j}$, which is $\mathcal{P}_t$-measurable. With known constant weights, every generator of $\mathcal{C}_t$ is a function of $(x_0, \dots, x_t)$, so $\mathcal{C}_t = \mathcal{P}_t$. Then $\mathbb{E}[x_{t+1} \mid \mathcal{P}_t] = ax_t + \kappa \sum_i q_i y_{i,t} = ax_t + \kappa \sum_m \big(\sum_i q_i \eta_i r_i^m\big) x_{t-m}$, because ε_{t+1} is independent of $\mathcal{P}_t$. □

Proof of (ii). With constant weights, x is an autoregression of infinite order, $x_t = \sum_{j \ge 1} \phi_j x_{t-j} + \varepsilon_t$, with $\phi_1 = a + \kappa k_0$, $\phi_j = \kappa k_{j-1}$ for $j \ge 2$, so $|\phi_j| \le \kappa W \bar{r}^{\,j-1}$ with $W = \sum_i w_i$. Since $\phi_j \ge 0$ and $\sum_j \phi_j = a + \kappa \sum_i q_i \le s < 1$, the polynomial $1 - \sum_j \phi_j z^j$ has no zero in $|z| \le 1$. For such processes, the same-realization mean squared prediction error of the least-squares autoregression of order L is $\sigma^2(1 + L/T) + \sigma^2 \sum_{j>L} O(\phi_j^2) + o(L/T)$ (Ing and Wei, 2003). The tail is $O(\bar{r}^{\,2L})$, so $L \asymp \log T$ gives $\sigma^2 + O(\sigma^2 \log T/T)$. The composition observer's error is at least σ^2, because ε_{t+1} is independent of $\mathcal{C}_t$. The constants depend on the process through $\bar{r}$, s and κW, which are bounded on the class. □

To check: Ing and Wei (2003) is stated for a fixed process. Uniformity over a compact stable class needs their constants to be bounded in terms of the spectral bounds, which holds here because the spectral density lies between $\sigma^2/(1+s)^2$ and $\sigma^2/(1-s)^2$.

Proof sketch of (iii). Within a window of m periods the kernel taps move by at most $|\Delta k_j| \le \eta_{\max} \bar{r}^{\,j} K v m$, so the autoregressive coefficients on the window lie within $O(vm)$ of their window average. Least squares on the window estimates that average with variance $O(L/m)$ in the forecast, and the drift adds a bias of order vm in coefficients, hence $O(v^2 m^2)$ in squared forecast error. Truncation adds $O(\bar{r}^{2L})$. The total $C\sigma^2(L/m + v^2 m^2 + \bar{r}^{2L})$ is minimized at $m \asymp v^{-2/3}$, $L \asymp \log(1/v)$. A complete proof follows the tracking analysis of Guo and Ljung (1995), which needs a persistence-of-excitation condition. Here that condition holds because the spectral density is bounded below. □

3 What breaks it

The theorem says composition is worthless for forecasting prices whenever the traders' flows are functions of public prices. Each way of breaking that assumption is a section of the paper or a prediction that can be tested.

Proposition 2 (The four exceptions)**.** *(a)* Informed classes. *If some group's flow depends on information not in $\mathcal{P}_t$, such as the next change in value, composition can have forecast value of order one.*

(b) Exogenous flows with different persistence. *If group i also trades an exogenous flow $u_{i,t}$, independent of prices, then in the linear-Gaussian case the forecast value of composition, beyond prices and the aggregate flow $A_t = \sum_i q_{i,t} y_{i,t-1} + \sum_i u_{i,t}$, equals κ^2 times the reduction in the error of forecasting $\sum_i u_{i,t+1}$ from the components $(u_{i,s})_{s \le t}$ instead of their sum. It is zero when all the u_i share one autoregressive law.*

(c) Position limits under stress. *With saturating rules, the next price depends on the population through a measure over memory and leverage that calm prices do not reveal (the paper's Theorem on saturation). Composition can then forecast prices once limits bind.*

(d) Targeted actions. *Composition is needed for actions aimed at one group, by the observation–intervention law. That concerns decisions, not forecasts.*

Proof. (a) is an example: let one group buy $\beta(\Delta f_{t+1})$, where f is the value. (b) Prices and aggregate flow determine the feedback parts, by (i). The remaining forecast error is κ times the error in forecasting $\sum_i u_{i,t+1}$, and Gaussian conditioning gives the stated equality. When all u_i follow one autoregression, so does their sum, and its own past is sufficient. (c) and (d) are results of the paper. □

Remark 3 (Why this matters)*.* The result has the structure of an irrelevance theorem: a benchmark under which something is irrelevant, and a list of exactly what makes it relevant, as in Modigliani and Miller (1958). The aggregation step is close in spirit to Gorman (1953). The prediction it makes is testable: class data should not improve price forecasts beyond prices and anonymous flow, except through (a)–(c). This may explain why trader-position reports have little forecasting power for futures returns (Sanders et al., 2009), which is to be checked against that literature before it is claimed.

4 Numerical checks

Check	Result	Script
(ii) spacing independence	excess MSE 1.56×10^{-4} for memories $(0.30, 0.60)$ and for $(0.44, 0.46)$	`irrelevance_theorem_check.py`
(iii) drift	excess$/v^{2/3}$ between 0.18 and 0.41 for v from $3 \cdot 10^{-5}$ to 10^{-3}	same
(b) aggregation channel	$\Delta R^2 \approx +0.1\%$ with persistences $0.3, 0.7, 0.95$; -0.03% with one persistence	`p1f_design_sim.py`
(a) informed class	$\Delta R^2 \approx +1.5\%$	same

The drift exponent fitted in simulation is 0.55, below 2/3 at small v because the window grid is capped. The bound in (iii) is an upper bound and is not shown to be tight.

5 For the proof check

1. Part (i): check that $\mathcal{C}_t = \mathcal{P}_t$ under known constant weights, including the initial condition.
2. Part (ii): read Ing and Wei (2003) and confirm that their conditions hold here (exponentially decaying coefficients; moment conditions on ε), and that their result is about same-realization prediction, as used.
3. Part (iii): write the full tracking proof, or state (iii) as a proposition with an explicit persistence-of-excitation assumption.
4. Proposition 2(b): verify the Gaussian conditioning step.
5. Novelty: search for prior forecast-irrelevance results for trader composition, and read Gorman (1953), Sanders et al. (2009).

References

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